13. TROUBLESHOOTING NETWORK SERVICES

Course: Diploma in Cyber Security

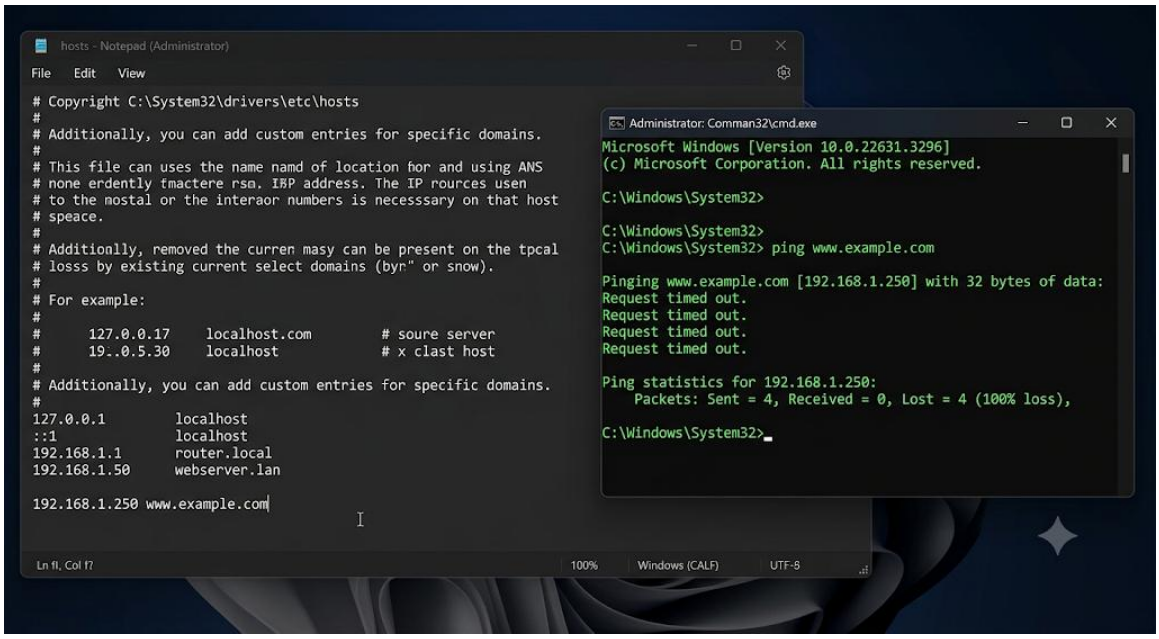
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Introduction

DNS and DHCP problems can make a network look like it has no internet even when the physical connection is fine. Troubleshooting should be done step by step instead of changing many settings at once.

Task 1: DNS failure with hosts file

In a controlled lab only, use a backup of the hosts file and create a temporary incorrect mapping for a test domain. Test the result and restore the original file immediately. Do not redirect real services or other users.



Task 2: Wrong DHCP gateway

In Packet Tracer/GNS3, configure an incorrect default gateway in the DHCP pool, connect a client, observe the wrong gateway, then correct the pool and renew the address.

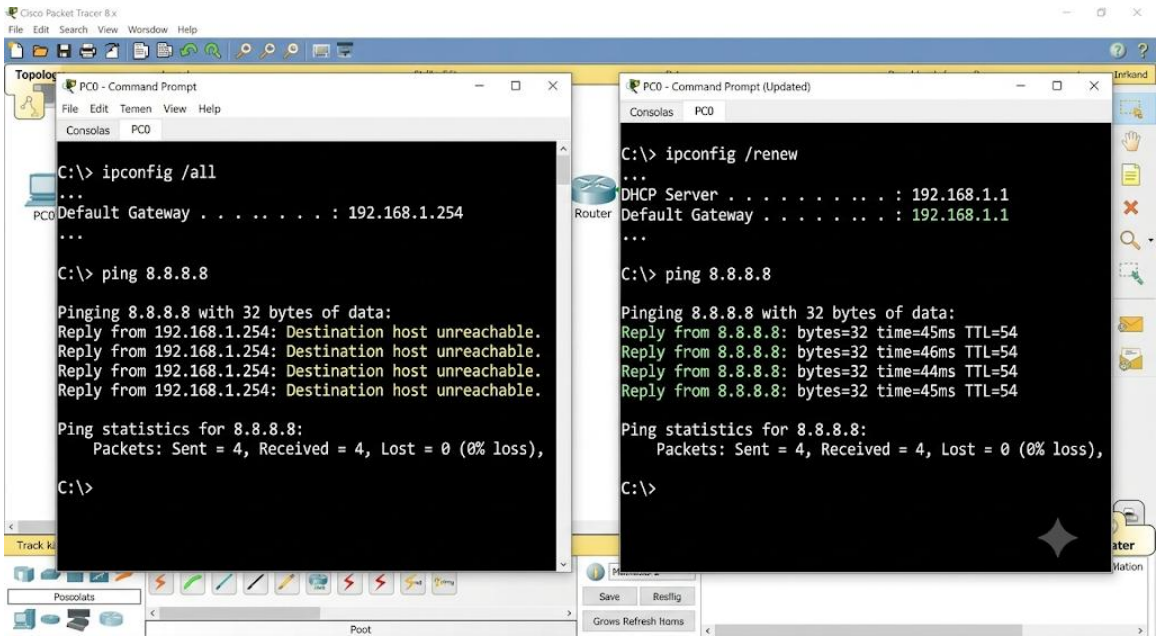


Figure 2: Before and After Comparison of DHCP Gateway Correction and IP Lease Renewal

Task 3: Website works by IP but not name

Use nslookup or dig. If name resolution fails while IP connectivity works, check the DNS server address, zone/record and client configuration.

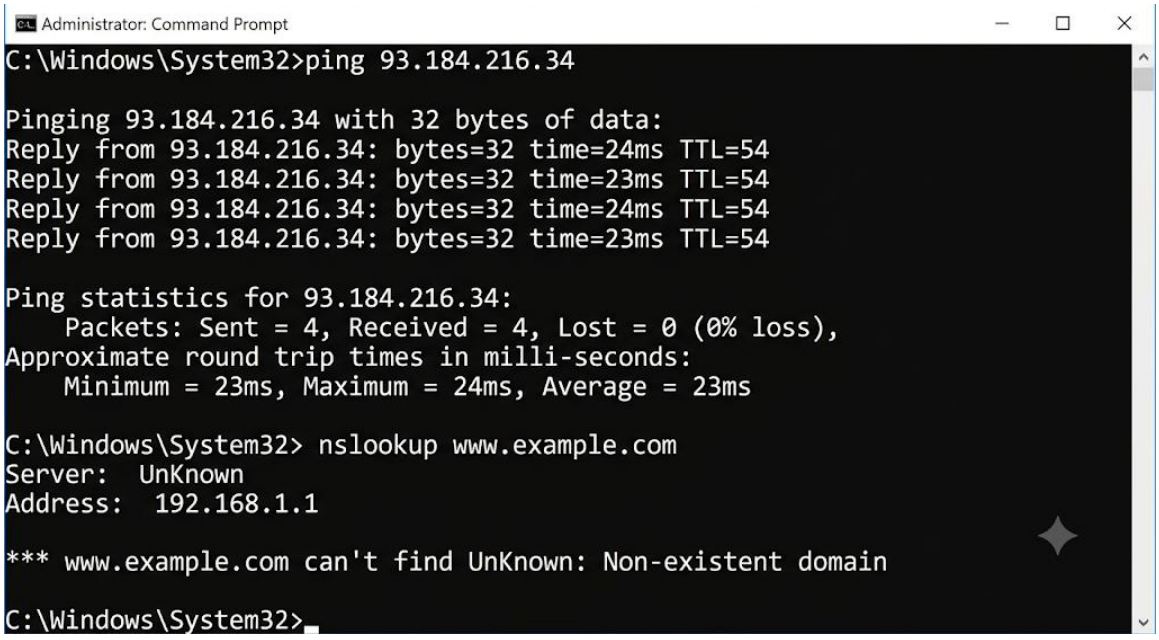


Figure 3: Diagnostic Output Demonstrating Successful Direct IP Reachability Alongside DNS Name Resolution Failure

Task 4: APIPA 169.x.x.x

Possible causes include DHCP server unavailable, exhausted scope, cable/Wi-Fi issue, VLAN mismatch or DHCP service failure. Check connectivity, scope availability and server status, then renew the address.

Diagnostic Step	Target Component	Corrective Action & Commands
Step 1: Physical Link Check	Network Cable / Wi-Fi Adapter	Verify link LEDs, ensure Wi-Fi profile is associated, and re-plug Ethernet patch cable.
Step 2: Service Verification	DHCP Server / Daemon Status	Ensure `DHCP Server` service (Windows) or `isc-dhcp-server` (Linux) is in `Running` state.
Step 3: Scope Pool Audit	Address Pool Capacity	Check server pool statistics to ensure available leases exist; expand subnet mask or reduce lease duration if exhausted.
Step 4: Client Address Renewal	Windows Client Network Stack	Run `ipconfig /release` followed by `ipconfig /renew` to force a new `DHCPDISCOVER` broadcast.

Task 5: DNS/DHCP issue table

Document at least three DNS issues and three DHCP issues with symptoms, possible causes and recommended fixes.

Service	Issue	Symptom	Possible Cause	Fix
DNS	Name does not resolve	Website name fails	Wrong DNS server/record	Check DNS settings and record
DNS	Wrong IP returned	Unexpected destination	Incorrect record/cache	Check record and clear cache
DNS	Server unreachable	nslookup times out	Network/service problem	Check connectivity and

				DNS service
DHCP	No IP address	169.x.x.x	DHCP unavailable	Check server/network
DHCP	Wrong gateway	Internet fails	Bad DHCP option	Correct gateway option
DHCP	No free address	Client cannot lease	Scope exhausted	Increase scope/release leases

Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.